

Noise-induced transitions in slow wave neuronal dynamics

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Abstract Many neuronal systems exhibit slow random alternations and sudden switches in activity states. Models with noisy relaxation dynamics (oscillatory, excitable or bistable) account for these temporal, slow wave, patterns and the fluctuations within states. The noise-induced transitions in a relaxation dynamics are analogous to escape by a particle in a slowly changing double-well potential. In this formalism, we obtain semi-analytically the first and second order statistical properties: the distributions of the slow process at the transitions and the temporal correlations of successive switching events. We find that the temporal correlations can be used to help distinguish among biophysical mechanisms for the slow negative feedback, such as divisive or subtractive. We develop our results in the context of models for cellular pacemaker neurons; they also apply to mean-field models for spontaneously active networks with slow wave dynamics.

Keywords Pacemaker neuron · Respiratory CPG · Noise-induced transitions · Noise effects · Bursting · Relaxation dynamics

1 Introduction

Noisy relaxation oscillator models for the random alternations of slow wave neuronal systems generate random

switching with fluctuations in active phase (*AP*) and silent phase (*SP*) durations. Noise can induce transitions even if the noise-free model is not oscillatory, but rather excitable or bi-stable (Baltanas and Casado 1998; Chow and White 1996; DeVille et al. 2005; Longtin et al. 1991; Rowat and Elson 2004; White et al. 1998). Analytical studies were done to characterize the statistical properties such as interspike interval distributions or durations of activity states in excitable cells (Chow and White 1996; Longtin et al. 1991) and in bursting neurons (Kuske and Baer 2002; Su et al. 2004; Hitczenko and Medvedev 2009).

In a model with relaxation dynamics, slow negative feedback sets the order of the time scale for *AP* and *SP* durations. It provides memory-like properties and can lead to correlations in successive *AP* and *SP* durations. Positive correlation between the *AP* and the previous *SP* was observed in several networks having relaxation dynamical features (Grzywacz and Sernagor 2000; Staley et al. 1998; Streit 1993; Tabak et al. 2001). But the correlation between the *AP* and the following *SP* was not found (Staley et al. 1998; Tabak et al. 2001). It has been suggested that this special pattern may be used to distinguish between different candidate mechanisms for the slow negative feedback (Tabak et al. 2001).

Interplay between the slow time dynamics and noise enables premature transitions between the activity states. A slower time scale of the negative feedback dynamics allows more time for noise to act, so even very small-amplitude noise can induce transitions. In a particular limit, as the amplitude of noise tends to zero and the time scale of the slow process increases to infinity accordingly, surprisingly very regular oscillatory behavior can emerge even in an excitable system (DeVill et al. 2005; Freidlin 2001; Muratov et al. 2005). In a spirit that parallels these previous works, we analyze the noise-induced transitions but without constraining the small noise and slow time scale to a

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particular relationship. (See also the extension (DeVillie et al. 2006) for the general case for small noise and slow time constant.) We obtain statistics for the slow process at the transitions and we extend the analysis to derive expressions and approximations for temporal correlations between successive transitions. It is shown that the relative variability and the speed of a slow process at transitions determine the temporal correlations.

We apply the analysis to two different biophysically-based, relaxation-like models of bursting neurons that are found in the subregion, called the pre-Bötzinger complex (preBötC) of the rat brain stem (Butera et al. 1999). The preBötC *in vitro* spontaneously generates rhythmic activity that is associated with the inspiratory phase of respiration. The pacemaker neurons in this region retain oscillatory bursting after blockade of synaptic transmission (Koshiya and Smith 1999). Different slow negative feedback mechanisms have been implicated as supporting the bursting and the contribution of the intrinsically bursting properties to the network rhythmic activity is controversial (Feldman and Del Negro 2006). Butera et al. (1999) developed two models with different negative feedback mechanisms each of which show oscillatory bursting behavior as observed in the experiments. We find, with an analysis of the noise induced transitions, that the different mechanisms induce different patterns of temporal correlations. We develop criteria, based on the dependence of these statistical properties on biophysical parameters, by which to distinguish among biophysical mechanisms, such as divisive or subtractive negative feedback. This framework can be extended to slow wave firing rate models of network dynamics.

2 Methods

We use single-compartment Hodgkin-Huxley type models with different negative feedback mechanisms formulated for the preBötC bursting pacemaker neurons (Butera et al. 1999). We eliminate spike-generating currents from the models so that they generate slow wave oscillations instead of oscillatory bursting.

Model 1 includes a slowly inactivating, instantaneously-activating persistent sodium current I_{NaP} and leakage current. The dynamics of voltage V and the slow inactivation gating variable h satisfy:

$$C \frac{dV}{dt} = -\bar{g}_{NaP} m_{P,\infty}(V) h (V - E_{Na}) - g_{Leak}(V - E_{Leak}) + I_{app} + \sigma' \xi \quad (1a)$$

$$\frac{dh}{dt} = \frac{h_\infty(V) - h}{\tau_h(V)} \quad (1b)$$

For a gating variable x , the steady-state voltage-dependent (in)activation function of x is $x_\infty(V) = \left(1 + \exp\left(\frac{V - \theta_x}{k_x}\right)\right)^{-1}$ and the voltage-dependent time constant is $\tau_x = \bar{\tau}_x / \cosh\left(\frac{V - \theta_x}{2k_x}\right)$. Parameters are given as in the paper (Butera et al. 1999): $C = 21$ pF, $\bar{g}_{NaP} = 2.8$ nS, $g_{Leak} = 2.8$ nS, $E_{Na} = 50$ mV, $\theta_{m_P} = -40$ mV, $k_{m_P} = -6$ mV, $\theta_h = -48$ mV, $k_h = -6$ mV and $\bar{\tau}_h = 10,000$ ms. The neuron's level of depolarization can be controlled by changing the mean applied current, I_{app} , which is equivalent to varying E_{Leak} . E_{Leak} and the amplitude of noise σ' are set to -59 mV and 10.5 pA respectively throughout the paper unless indicated otherwise.

Model 2 has a leakage current, a fast persistent sodium current I_{NaP} that does not inactivate (*i.e.*, $h \equiv 1$), and incorporates a slowly activating K^+ current, I_{KS} . The equations for V and w , the slow activation gating variable for I_{KS} satisfy:

$$C \frac{dV}{dt} = -\bar{g}_{NaP} m_{P,\infty}(V)(V - E_{Na}) - \bar{g}_{KS} w (V - E_K) - g_{Leak}(V - E_{Leak}) + I_{app} + \sigma' \xi \quad (2a)$$

$$\frac{dw}{dt} = \frac{w_\infty(V) - w}{\tau_w(V)} \quad (2b)$$

The common parameters of *model 1* and *2* have identical values and $\bar{g}_{KS} = 5.6$ nS, $E_w = -85$ mV, $\theta_w = -38$ mV, $k_w = -6$ mV and $\bar{\tau}_w = 10,000$ ms.

We define the transition between high and low voltage states using voltage criteria: -42 mV (-40 mV) from *SP* to *AP*, -44 mV (-45 mV) from *AP* to *SP* in *model 1* (*model 2*). If transitions happen less than 1,000 times or it is hard to define transitions due to noise, we treat it as no transition. The dependence of the number of transitions on E_{Leak} and σ' will be discussed more in Section 3.4.

These stochastic differential equations are integrated numerically over 10^5 seconds using the Euler-Maruyama method (Higham 2001) implemented in the C programming language. With $E_{Leak} = -59$ mV and $\sigma' = 10.5$ pA, each model generates more than 10^5 transitions. White noise is generated using the Box-Muller method (Box and Muller 1958). The integration time step was set to 0.01 ms. All statistical properties were preserved when the time step was halved. Matlab is used to compute statistical quantities and to realize graphics.

3 Results

We consider single-compartment Hodgkin-Huxley type models that were developed to describe the activity of

bursting neurons in the preBötC (Butera et al. 1999). These neurons burst in their role during the inspiratory phase of respiration. They generate bursting on a time scale of seconds. Slow negative feedback having a time constant of the same order is necessary. Here, we examine two different versions of the model, corresponding to two different negative feedback mechanisms: one with slow inactivation of an inward current I_{NaP} , which is denoted by *model 1* and the other with slow activation of an outward current I_{KS} , denoted by *model 2*.

We consider the effect of noise on the statistical properties of bursting for two different underlying mechanisms. Fluctuations in voltage are generated by adding white noise to the instantaneous external current. We disable spike-generating currents to obtain idealized slow wave systems. These reduced two-variable models can be either oscillatory or excitable.

3.1 The deterministic dynamics

The two models of preBötC bursting neurons have common dynamical features. Without spike-generating currents, the models generate slow wave oscillations in voltage, V , via slow negative feedback. The time scales of V (fast) and the negative feedback (very slow) are sufficiently different that the noise free dynamics have relaxation character. We illustrate some dynamical features using *model 1*. The time courses of V and the slow negative feedback h , and the trajectory in the V - h phase plane are shown in Fig. 1(a, 1–2). For some range of applied current, I_{app} , the model shows periodic activity. It corresponds to a limit cycle whose fast and slow flows are readily identifiable. Due to the strong separation of time scales, the motion on the limit cycle is broken up into fast and slow phases. During the slow phases, V gradually increases at a low voltage value, denoted as the silent phase (*SP*) and decreases at a high voltage value, denoted as the active phase (*AP*). The phases terminate with sudden jumps between the states (Fig. 1(a, 1)). In a more complete version of the model, the *AP* depolarization leads to spiking, hence providing the bursting profile. In the phase plane, the *SP* and *AP* correspond to drift along the slow manifolds which are the attracting branches of the V -nullcline. A transition between states occurs, in the noise-free case, when the trajectory encounters or falls off a knee of the slow manifold (Fig. 1(a, 2)).

When a sufficiently strong transient stimulus is introduced after time a during the *SP* in a noise-free case (arrows in Fig. 1(a, 1–2)), the voltage jumps to a depolarized value and initiates a new *AP*. Duration of the ensuing *AP*, denoted by b is shorter than that without a perturbing stimulus. Since the slow negative feedback was not fully recovered during the previous *SP*, it takes a shorter time for the negative feedback to reach the level to terminate the *AP* and the trajectory

reaches the knee of the right branch sooner. For an earlier perturbing stimulus the *SP* shortens, i.e., a decreases, this induces a shorter *AP* (Fig. 1(a, 3)).

Rapid fluctuations reflect the noisy current variations (Fig. 1(b, 1)); correspondingly, the trajectory in the phase plane shows horizontal spread along the left and right branches (Fig. 1(b, 2)) as the trajectory varies cycle-to-cycle. Noisy relaxation oscillator models for such systems generate random switching with fluctuations in *AP* and *SP* durations. Noise generates premature transitions between the states. Transitions between the *AP* and *SP* occur before the trajectory reaches a knee. Even if the noise-free model is not oscillatory, but rather excitable or bi-stable, transitions can be induced. The presence of noise means that we cannot predict exactly the next *AP* duration from the current *SP* or next *SP* duration, but can we describe the durations and switching points statistically?

3.2 Analysis of noise-induced transitions in fast–slow dynamics

Here we analyze the noise-induced (premature) transitions of fast–slow dynamics in this neuronal model and obtain transition probability density functions (*PDFs*) for the slow variable at the transition points. Based on this analysis, we also obtain distributions of durations of each state. See also the treatment (DeVillie et al. 2006) where the mean and variance are derived in a similar context.

Due to the strong separation of time scales, the system can be dissected into fast and slow subsystems. Since noise is presented only in the dynamics of the fast variable, the dynamics of the fast subsystem is described by a stochastic differential equation. First, we formulate this equation in terms of a potential function, $U(V, h)$. This potential is obtained by integrating with respect to V the model's instantaneous current-voltage relation, denoted as $I_{inst}(V, h)$. The dynamics of *model 1*, after expressing the velocity of h as $g(V, h)$, is rewritten as follows:

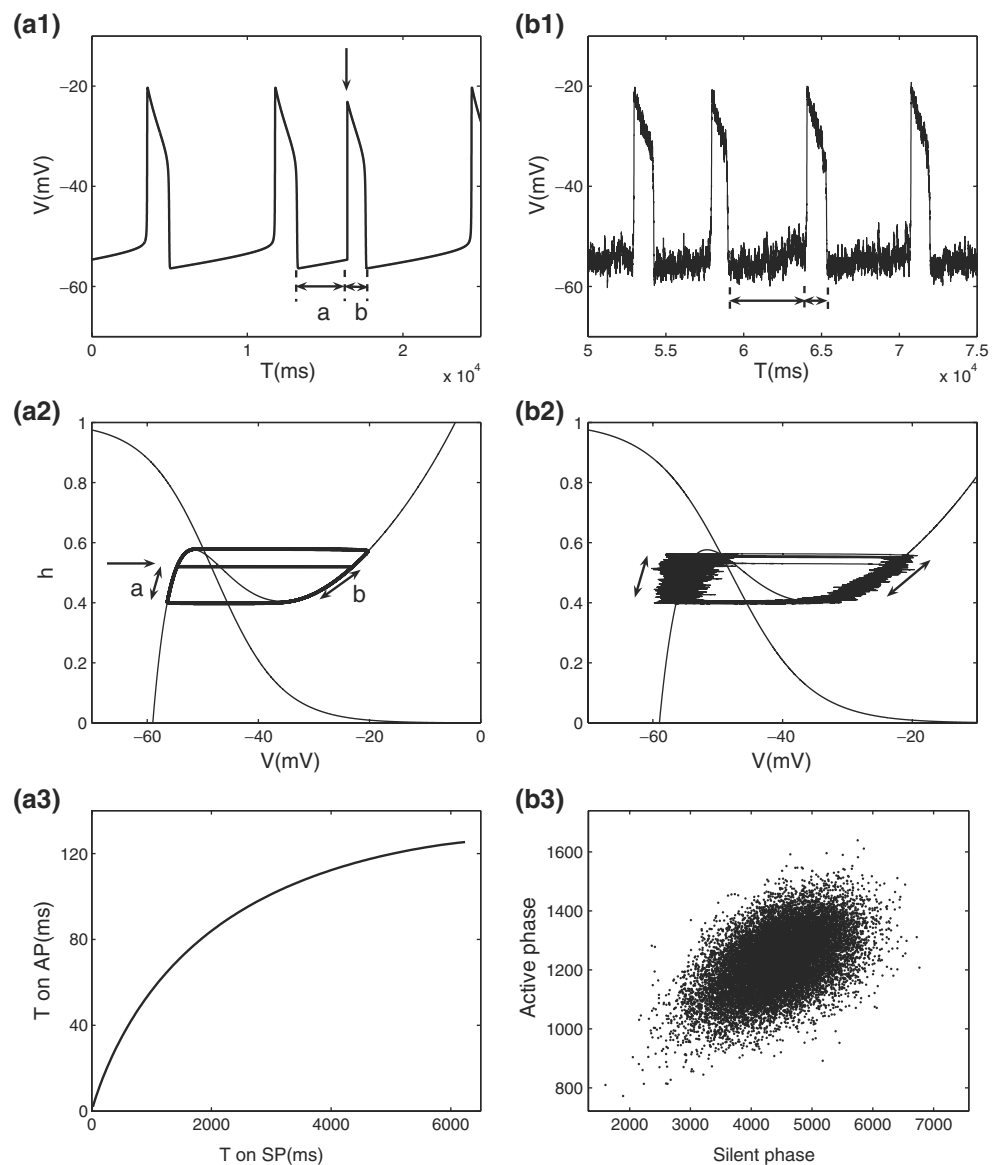
$$U(V, h) = \int \frac{I_{inst}(V, h)}{C} dV \quad (3a)$$

$$dV = -\frac{I_{inst}(V, h)}{C} dt + \frac{\sigma'}{C} dW_t = -\frac{\partial U(V, h)}{\partial V} dt + \sigma dW_t \quad (3b)$$

$$dh = g(V, h) dt \quad (3c)$$

Figure 2 illustrates the fast–slow dynamics in relation to the potential function. At each h , a potential is given and the local minima and maxima are zeros of the current-voltage relation since local minima and maxima satisfy $\partial U(V, h)/\partial V = I_{inst}(V, h) = 0$. In the V - h phase plane, the

Fig. 1 Dynamics of *model 1* depending on the presence of noise. *Left: (a, 1–3) without noise. Right: (b, 1–3) with noise.* Panel 1 and 2 show the time trace of voltage and the trajectory in V - h phase plane. Arrows in *a1* and *a2* indicate that a transient stimulus is introduced to generate a premature transition from the *SP* to start a new *AP*. The next *AP* is longer if the switch is induced later in the *SP* (*a3*). In a noisy oscillator, we can observe vertical fluctuations in the voltage time course (*b1*) and horizontal spread of V in the phase plane (*b2*). In this case with noise, *SP* and *AP* durations vary randomly. For *model 1* the *SP* duration and next *AP* duration are correlated (*b3*), as in the noise-free case



V -values on the cubic-shaped V -nullcline are steady states of the fast subsystem, so they correspond to local minima or maxima of the potential at given h . In the phase plane (Fig. 2(b)), a horizontal line with constant h has three intersections with the V -nullcline, indicated by asterisks. Intersections on the left and right branches are stable; they correspond to local minima of the potential. The middle point is a local maximum. The profile of $U(V, h)$ versus V is that of a double-well potential whose shape depends on h (Fig. 2(c)).

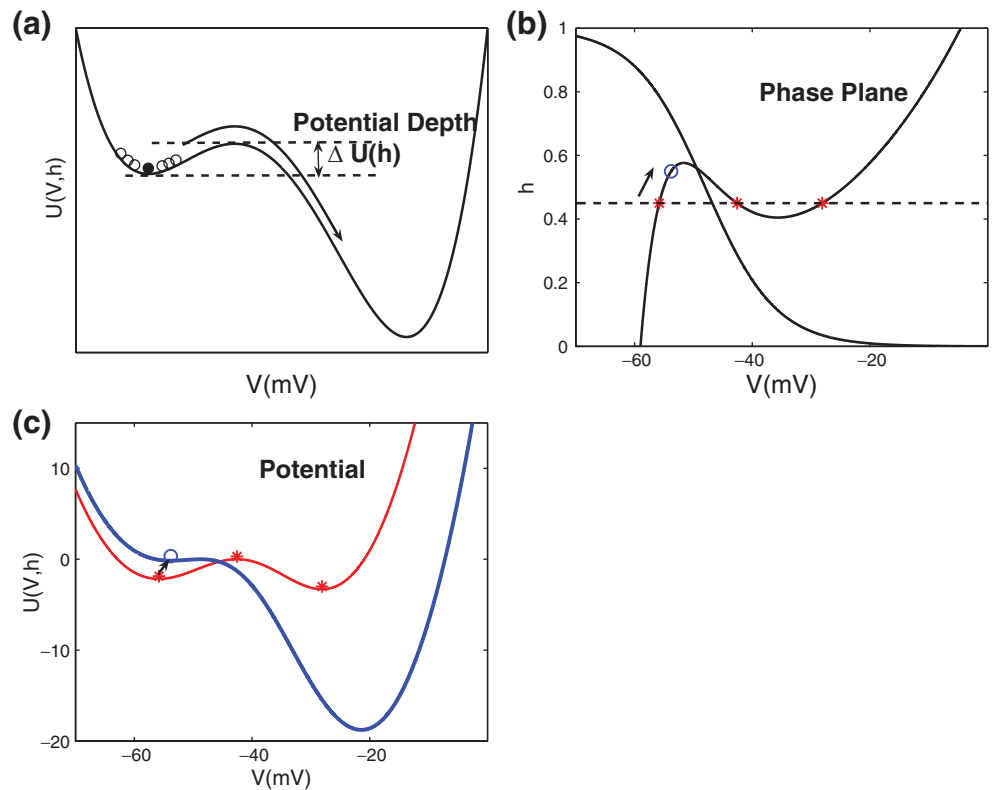
When the potential is a double-well and white noise is presented, noise-induced transitions in a double-well potential are illustrated in Fig. 2(a). Without noise, the system stays at one of local minima of $U(V, h)$ and weak noise leads to small perturbations around it. A transition from one well to the other occurs as a jump over the potential barrier. In the case of white noise with small

amplitude, the mean escape rate from one well to the other is given by Kramers rate, λ_K and this rate is dependent on the wells, denoted by ΔU and on the noise amplitude, σ (Hänggi et al. 1990; Gardiner 1985). For each h , the potential depth of a well is the potential difference between a local maximum and a local minimum, and λ_K as a function of h is given as follows:

$$\lambda_K(h) \approx \nu(h) \exp(-2\Delta U(h)/\sigma^2) \quad (4)$$

Now let's take the dynamics of the slow variable into account. With the dynamics of h , the system slowly moves up along the V -nullcline during the *SP* (circle in Fig. 2(b)). As it gets closer to the knee, the left potential well gets shallower. Since the height of the potential barrier gets smaller, a transition from the left well to the

Fig. 2 Fast–slow dynamics in terms of the potential function $U(V, h)$ of the fast subsystem. **(a)** Schematics of the double-well potential for fixed slow variable, h . Escape from one well over the barrier is analogous to the transition from one stable state to the other. **(b)** and **(c)** Slow motion along the V -nullcline during the SP and change of the potential function. Curves in **c** are potentials at different h . As h increases slowly, the left well of potential becomes more shallow so that the transition happens more easily



right well happens with greater likelihood. This transition is a horizontal jump from the left branch of the V -nullcline to the right branch in the phase plane and a new AP starts.

We can describe the transitions in the fast–slow dynamics and obtain the transition PDF s from SP to AP , denoted as $\rho_{SP \rightarrow AP}$ and from AP and SP , denoted as $\rho_{AP \rightarrow SP}$ by using Kramers rate and survival analysis (Cox and Lewis 1966; Elandt-Johnson and Johnson 1980). The transition from one state to the other corresponds to the event or death in a survival analysis. Let T denote the time of an escaping event given a starting point at some h , then Kramers rate, λ_K leads to a hazard function, λ which is the event rate at time t conditional on survival until t . During the AP and SP , h changes monotonically, so we can replace t with h using the dynamics of h . So a survival function S from its definition and the transition PDF ρ as a function of h are given as follows:

$$S(t) = \Pr(T > t) \quad (5a)$$

$$\lambda(t)dt = \Pr(t \leq T \leq t + dt | T > t) = \frac{-S'(t)dt}{S(t)} \quad (5b)$$

$$\frac{dS}{dh} \frac{dh}{dt} = -S(h)\lambda(h) = -S(h)\lambda_K(h) \quad (5c)$$

After dividing Eq. (5c) by $S(h)$ and $\frac{dh}{dt}$ and then integrating, we obtain Eq. (5d).

$$\begin{aligned} S(h) &= \exp\left(-\int \frac{\lambda_K(h')}{g(V(h'), h')} dh'\right) \\ &= \exp\left(-\int \frac{v(h') \exp(-2\Delta U(h')/\sigma^2)}{g(V(h'), h')} dh'\right) \end{aligned} \quad (5d)$$

From Eq. (5d), we get the transition PDF $\rho(h)$.

$$\begin{aligned} \rho(h) &= -\frac{dS(h)}{dh} = \frac{v(h) \exp(-2\Delta U(h)/\sigma^2)}{g(V(h), h)} \\ &\quad \exp\left(-\int \frac{v(h') \exp(-2\Delta U(h')/\sigma^2)}{g(V(h'), h')} dh'\right) \end{aligned} \quad (5e)$$

From the definition of survival function S and hazard function λ , we obtain the relation of two functions described in Eq. (5b). In Eq. (5c), we expressed those functions as a function of the slow variable, h and the Kramers rate, λ_K replaces λ . In Eq. (5d), we assume that S is always 1 when a new state begins. That is, there is no escape at the beginning of a SP or AP . This assumption is a good approximation as long as the two knees of the V -nullcline are far apart and the potential well is deep enough that the escape rate λ_K is almost 0 when a new state begins. Fig. 3(a, b) show the time courses of V and h and

trajectories in the phase plane. The level of h at transitions varies every cycle (3a), and the transition $PDFs$ in h , $\rho_{SP \rightarrow AP}$ and $\rho_{AP \rightarrow SP}$ are obtained from simulation (3b). We can see that the histograms for the two transition $PDFs$ are well separated and there is no transition at the beginning of each state. Therefore we may apply the analysis: evaluate $\rho_{SP \rightarrow AP}$ using Eq. (5e), and plot the $PDFs$ as heavy curves in Fig. 3(c). They agree well with the histograms for the transitions obtained from the simulations. The analysis can be applied even when the system is not oscillatory.

When the left or right branch of the V -nullcline has an intersection with the h -nullcline of the slow variable and the system gets attracted to that point, then the velocity of the slow variable becomes 0. That is, if $g(V(h), h)$ becomes zero in Eq. (5d), the analysis breaks down and we cannot get the transition $PDFs$. However, if the noise is small but sufficient, it induces a transition before the attractor is reached, then the analysis still holds. The non-oscillatory case will be discussed again in Section 3.4.

Transition $PDFs$ give information about where the system enters and leaves a state. For example, a SP starts with $\rho_{AP \rightarrow SP}$ and ends with $\rho_{SP \rightarrow AP}$. We can now use this information to obtain distributions of duration in each state, denoted as T_{SP} and T_{AP} . Assuming that the system moves along the V -nullcline, we obtain the speed of motion on the slow manifolds, that is, $g(V(h), h)$ in Eq. (5d). Then the duration is the integral of $1/g(V(h), h)$. If the location of leaving the state is not dependent on where it starts, then the distribution of durations can be obtained directly from

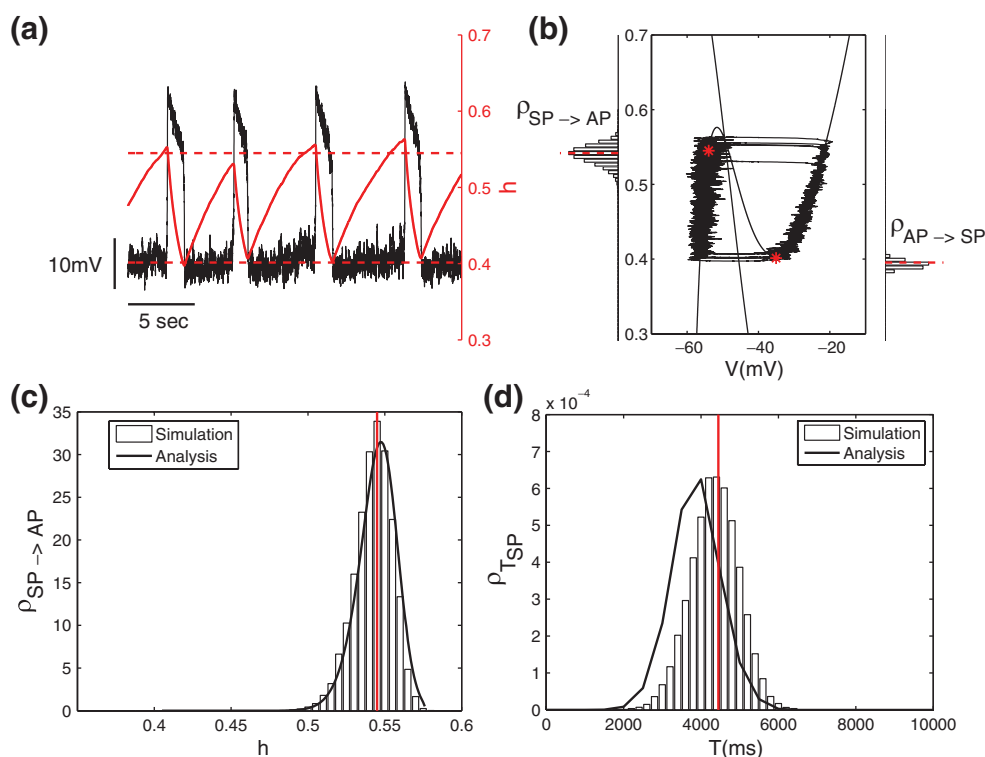
the transition $PDFs$. As long as the transition $PDFs$ are well separated, we can assume independence, since due to strong separation of time scales, the trajectory is reset and moves along the V -nullcline no matter where it starts and the memory of the starting location is lost during the slow motion. We checked the assumption of independence: distributions of durations computed with distributions of pair-wise starting and leaving locations obtained from the simulation are the same as those under the independent assumption. The computed distribution of SP durations is compared with the histogram of durations obtained from the simulation in Fig. 3(d). The shapes of the distributions are similar but the analysis underestimates the durations. The discrepancy between the analysis and the simulation is discussed in the “Appendix”.

3.3 Temporal correlation between durations in SP and AP

Next we apply and extend the analysis of noise-induced transitions to characterize some second order statistical properties such as temporal correlation between successive switching events. We apply the analysis to two models that have different biophysical mechanisms for slow wave generation. In Section 3.4, we will discuss how the underlying biophysical mechanism affects statistical properties. Here, we focus on the relation between the transition $PDFs$ and the temporal correlation.

The slow wave oscillation in *model 1* is modulated by the inactivation of an inward current and by activation of an

Fig. 3 Noise-induced transitions in fast-slow dynamics. (a) and (b): Time courses (a) and trajectories in the V - h phase plane (b) in the presence of noise. The h values at the transitions fluctuate around the mean which is indicated by dashed horizontal lines and dots; histograms for h -values in b for transitions were obtained from simulation. (c) and (d): Transition $PDFs$ and distribution of SP durations obtained from simulation and analysis. Good agreement of analysis (solid curves) with simulation is seen in (c); modest discrepancies in (d) between the simulation (histogram of SP durations) and the analysis (solid curves) are discussed in the “Appendix”



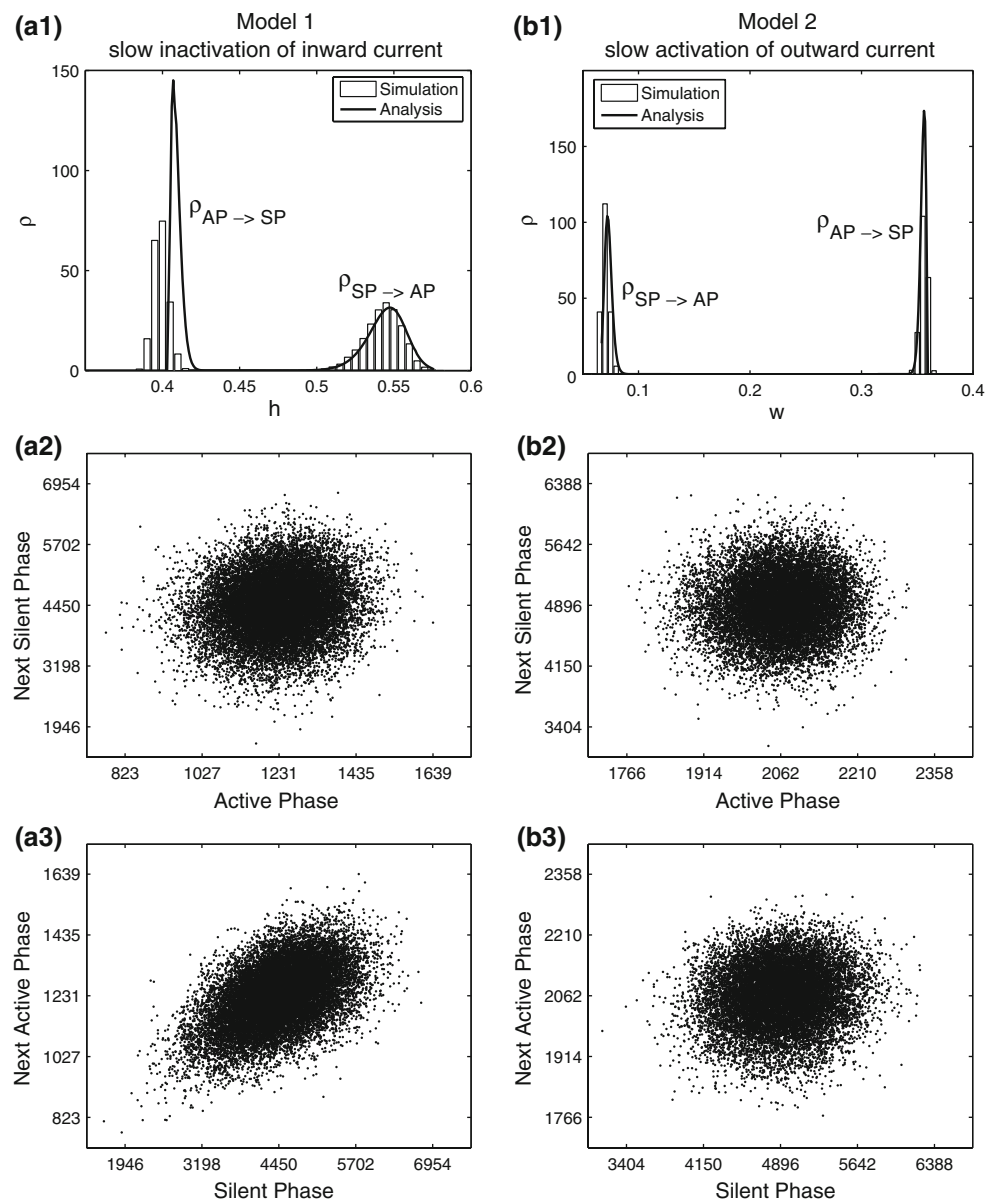
outward current in *model 2*. In Section 3.1, we described temporal correlations in a deterministic case: the shorter is the current state, the shorter will be the following state. Thus, successive durations are predictably correlated (Fig. 1(c)). In a noisy relaxation oscillator, we still expect some correlation since a slow process provides a memory-like property. However, noticeable correlation is not necessarily observed even if the noise is small.

Figure 4 illustrates differences in statistical properties (transition PDFs and temporal correlations) of two models obtained from simulations over long time intervals. Simulation and analysis show that $\rho_{SP \rightarrow AP}$ is flatter in *model 1* (Fig. 4(a, 1)), while the variances of $\rho_{SP \rightarrow AP}$ and $\rho_{AP \rightarrow SP}$ are similar in *model 2* (Fig. 4(b, 1)). The temporal correlations of successive durations such as the durations of *SP* and following *AP* or *AP* and following *SP* are computed from

the simulation. Simulation with noise shows that only the *SP* and next *AP* in *model 1* are temporally correlated (Fig. 4(a, 3)), but others are not (Fig. 4(a2, b2 and b3)).

The two models have similar distributions of durations in the *SP* and *AP* (data not shown). With $E_{Leak} = -59$ mV, the *SP* is closer to the *h* or *w*-nullcline than the *AP* in both models. This leads to slower motion in the *SP* than that in the *AP* and the mean and variance of T_{SP} are larger than those of T_{AP} . So the distributions of T_{SP} and T_{AP} don't show significant difference in the two models, while temporal correlations do: they are asymmetric in *model 1* and symmetric in *model 2*. Thus, the temporal correlations cannot be inferred only from the distributions of durations. Instead, we observe that *model 1* and *2* show a difference in statistical properties in slow variables such as variances of transition PDFs, so it is reasonable to hypothesize that

Fig. 4 Comparison of statistical properties of switching events for *model 1* (left panels) and *model 2* (right panels). Panels *a1* and *b1* show histograms (from simulation) and PDFs (analysis) for transition *h* or *w*-values. In *model 1*, $\rho_{SP \rightarrow AP}$ is much broader than $\rho_{AP \rightarrow SP}$, while in *model 2* the distributions have similar variance (*a1* and *b1*). Panels in rows 2 and 3 show the temporal correlations of successive states. Little if any temporal correlations are found for *AP* to next *SP* (*a2*, *b2*) and for *SP* to next *AP* in *model 2* (*b3*). Only one type of transition (*SP* to *AP* in *model 1*) shows substantial correlation (*a3*)



significantly different temporal correlations are involved with the asymmetrical variances of transition *PDFs* for the slow variables.

To verify the hypothesis, let's calculate the temporal correlations. Temporal correlations between successive switching events can be computed with transition *PDFs* and the slow dynamics as we obtained the distributions of durations in Section 3.2. Under the assumption that the fast fluctuation in the V along the slow manifold is averaged out in the slow dynamics, the speed of motion in the *SP* and *AP* for each h becomes the speed of motion on the slow manifolds, that is, $g(V(h), h)$. Let's denote $c_{SP}(h)$ and $c_{AP}(h)$ as the speed during the *SP* and *AP*. With the speed of motion and the transition *PDFs*, we can derive expressions for the temporal correlation between *SP* and next *AP* from the transition *PDFs*. In Fig. 5(a), the *SP* and next *AP* share the level of h at which *SP* ends and the next *AP* starts, so let's denote X_1 as h where *SP* starts, Y as h where *SP* ends and next *AP* starts and X_2 as h where next *AP* ends and new *SP* starts. Then the duration of each state is expressed as an integral.

$$T_{SP} = \int_{X_1}^Y \frac{1}{c_{SP}(h')} dh' = \int_{X_1}^{\mu_X} \frac{1}{c_{SP}(h')} dh' + \int_{\mu_X}^{\mu_Y} \frac{1}{c_{SP}(h')} dh' + \int_{\mu_Y}^Y \frac{1}{c_{SP}(h')} dh' \quad (6a)$$

$$T_{nextAP} = \int_{X_2}^Y \frac{1}{c_{AP}(h')} dh' = \int_{X_2}^{\mu_X} \frac{1}{c_{AP}(h')} dh' + \int_{\mu_X}^{\mu_Y} \frac{1}{c_{AP}(h')} dh' + \int_{\mu_Y}^Y \frac{1}{c_{AP}(h')} dh' \quad (6b)$$

To see the relation of transition *PDFs* and temporal correlations explicitly, we make a few approximations. First, we approximate the transition *PDFs* with Gaussian normal distributions, $n(\mu, \sigma^2)$ where μ and σ^2 are the mean and variance of the transition *PDF*. Then the distributions of $X_{1,2}$ and Y follow the transition *PDFs*, that is, $X_{1,2} \sim \rho_{AP \rightarrow SP} \approx n(\mu_{AP \rightarrow SP}, \sigma_{AP \rightarrow SP}^2)$ and $Y \sim \rho_{SP \rightarrow AP} \approx n(\mu_{SP \rightarrow AP}, \sigma_{SP \rightarrow AP}^2)$. Next, we assume that the flow near the transitions is at constant speed so that the speeds of motion near X_1 and Y in the *SP* are denoted as c_{SP}^I and c_{SP}^T . Similarly, c_{AP}^I and c_{AP}^T denote the speeds near Y and X_2 in the *AP*. Here, I and T stand for the initiation and termination of the state. The constant speed of motion near transitions is a good approximation when the noise is of small-amplitude, since then the h -values at the transitions as well as the speed of motion vary less. Also the assumption provides a better estimate away from the case in which the speed changes considerably near the transitions. It will be discussed more in Section 3.4.

Then T_{SP} and T_{nextAP} are given as follows.

$$T_{SP} = \int_{X_1}^Y \frac{1}{c_{SP}(h')} dh' = \frac{\mu_X - X_1}{c_{SP}^I} + \int_{\mu_X}^{\mu_Y} \frac{1}{c_{SP}(h')} dh' + \frac{\mu_Y - Y}{c_{SP}^T} \quad (6c)$$

$$T_{nextAP} = \int_{X_2}^Y \frac{1}{c_{AP}(h')} dh' = \frac{\mu_X - X_2}{c_{AP}^T} + \int_{\mu_X}^{\mu_Y} \frac{1}{c_{AP}(h')} dh' + \frac{\mu_Y - Y}{c_{AP}^I} \quad (6d)$$

In these expressions, the second integral terms from μ_X to μ_Y are fixed in T_{SP} and T_{nextAP} and they become the mean T_{SP} and T_{nextAP} . Thus, the temporal correlation satisfies

$$\begin{aligned} \text{Cor}(T_{SP}, T_{nextAP}) &= \frac{E((T_{SP} - \mu_{T_{SP}})(T_{nextAP} - \mu_{T_{nextAP}}))}{\sqrt{\text{var}(T_{SP})\text{var}(T_{nextAP})}} \\ &= \frac{E\left(\left(\frac{\mu_X - X_1}{c_{SP}^I} + \frac{\mu_Y - Y}{c_{SP}^T}\right)\left(\frac{\mu_X - X_2}{c_{AP}^T} + \frac{\mu_Y - Y}{c_{AP}^I}\right)\right)}{\sqrt{\text{var}\left(\frac{\mu_X - X_1}{c_{SP}^I} + \frac{\mu_Y - Y}{c_{SP}^T}\right)\text{var}\left(\frac{\mu_X - X_2}{c_{AP}^T} + \frac{\mu_Y - Y}{c_{AP}^I}\right)}} \\ &= \frac{\frac{\text{var}(Y)}{c_{SP}^I c_{AP}^I}}{\sqrt{\left(\frac{\text{var}(X_1)}{c_{SP}^I{}^2} + \frac{\text{var}(Y)}{c_{SP}^T{}^2}\right)\left(\frac{\text{var}(X_2)}{c_{AP}^T{}^2} + \frac{\text{var}(Y)}{c_{AP}^I{}^2}\right)}} \\ &= \frac{\frac{c_{SP}^I}{c_{SP}^T} \text{var}(Y)}{\sqrt{(\text{var}(X_1) + (\frac{c_{SP}^I}{c_{SP}^T})^2 \text{var}(Y))((\frac{c_{AP}^I}{c_{AP}^T})^2 \text{var}(X_2) + \text{var}(Y))}} \\ &= \frac{\frac{c_{SP}^I}{c_{SP}^T} \sigma_{SP \rightarrow AP}^2}{\sqrt{\left(\sigma_{AP \rightarrow SP}^2 + \left(\frac{c_{SP}^I}{c_{SP}^T}\right)^2 \sigma_{SP \rightarrow AP}^2\right)\left(\left(\frac{c_{AP}^I}{c_{AP}^T}\right)^2 \sigma_{AP \rightarrow SP}^2 + \sigma_{SP \rightarrow AP}^2\right)}} \end{aligned} \quad (6e)$$

Similarly, the temporal correlation of AP and next SP is given as follows.

$$Cor(T_{AP}, T_{nextSP}) = \frac{\frac{c_{AP}^J}{c_{AP}^I} \sigma_{AP \rightarrow SP}^2}{\sqrt{\left(\sigma_{AP \rightarrow SP}^2 + \left(\frac{c_{SP}^J}{c_{SP}^I} \right)^2 \sigma_{SP \rightarrow AP}^2 \right) \left(\left(\frac{c_{AP}^J}{c_{AP}^I} \right)^2 \sigma_{AP \rightarrow SP}^2 + \sigma_{SP \rightarrow AP}^2 \right)}} \quad (6f)$$

If the flow is at constant speed during the entire SP and AP , respectively, then $\frac{c_{SP}^J}{c_{SP}^I}$ and $\frac{c_{AP}^J}{c_{AP}^I}$ become 1. In Eq. (6e) and (6f), the temporal correlations turn into a fractional ratio of variances of transition $PDFs$ such as

$$Cor(T_{SP}, T_{nextAP}) = \frac{\sigma_{SP \rightarrow AP}^2}{\sigma_{AP \rightarrow SP}^2 + \sigma_{SP \rightarrow AP}^2} \quad (6g)$$

$$Cor(T_{AP}, T_{nextSP}) = \frac{\sigma_{AP \rightarrow SP}^2}{\sigma_{AP \rightarrow SP}^2 + \sigma_{SP \rightarrow AP}^2} \quad (6f)$$

The sum of the two temporal correlations is 1. In *model 1*, $\text{var}(\rho_{SP \rightarrow AP}) \gg \text{var}(\rho_{AP \rightarrow SP})$, so we expect large correlation between T_{SP} and T_{nextAP} and small correlation between T_{AP} and T_{nextSP} . And in *model 2* with similar variances, the correlations would be similar, close to one-half. Our approximate analysis agrees with our claim that asymmetry (symmetry) in variances of the transition $PDFs$ induces asymmetry (symmetry) in the temporal correlation.

However, the values of temporal correlations under the assumption of constant speed during the entire state differ from those obtained from the simulations. The variance of $\rho_{SP \rightarrow AP}$ is almost ten times larger than the variance of $\rho_{AP \rightarrow SP}$ in *model 1*, so the temporal correlation between SP and next AP is predicted to be close to 1, while it is close to 0.5 in simulations. Similarly, temporal correlations in *model 2* from the analysis are close to 0.5 but those from simulation are close to 0.1. The sum of the two temporal correlations is considerably less than 1.

The relative speeds of the slow variable in the SP and AP should be taken into account in order to predict the values of temporal correlations. In Eq. (6e) and (6f), the variabilities of the terminal points, $\sigma_{SP \rightarrow AP}^2$ in the SP and $\sigma_{AP \rightarrow SP}^2$ in the AP are multiplied by the ratio of the speed at the entry points to the speed at the terminal points, $\frac{c_{SP}^J}{c_{SP}^I}$ and $\frac{c_{AP}^J}{c_{AP}^I}$, respectively. Therefore, these ratios of speeds can be interpreted as amplification factors of the variability of the terminal points to the variability of entry points when the speed is not constant during the entire state. For instance, under the assumption of constant speed, the temporal

correlation from SP to AP is almost 1 since in Fig. 5(a), sharing Y which is broadly distributed, accounts for most of the variability. The SP and next AP have only a little uncertainty in X_1 and X_2 . However, in the simulation, the system slows down as it drifts toward the knees because it gets close to the h or w -nullcline (Fig. 5(b)) and it spends relatively more of the AP time near X_2 than near Y during the AP . Hence, the variability of X_2 is amplified in the AP durations by a factor of $\frac{c_{AP}^J}{c_{AP}^I}$. With large $\frac{c_{AP}^J}{c_{AP}^I}$, the denominator increases in Eq. (6e) and the temporal correlation of the SP and next AP is reduced from that fractional ratio of transition $PDFs$. Still, the temporal correlations in model 1 are different but not as much as we expected from the analysis assuming the constant speed of motion in the entire state.

3.4 Relationship of temporal correlations to different slow wave mechanisms

In Section 3.3, we described how the temporal correlations between successive switching events depend on the transition $PDFs$ and the relative speeds of the slow variable during the AP and SP by deriving the semi-analytical expressions of the temporal correlations. We applied the analysis to two different models of a preBötC pacemaker neuron and depending upon the mechanisms for the slow negative feedback, the system has symmetric or asymmetrical transition $PDFs$. Here we investigate how the underlying mechanism affects the statistical properties. And we further ask whether these properties can be used to distinguish the mechanisms of the slow wave generation.

In order to understand frequency control of respiration we can ask about neuronal mechanisms for frequency control in the central pattern generator circuits. In simulations E_{Leak} has been used as a control parameter of the oscillation period; its affect on period is equivalent to changing the net input, as modeled by I_{app} (Butera et al. 1999). Figure 6 illustrates how the temporal correlations of the two models depend on E_{Leak} . As E_{Leak} increases, both models progress through regimes of quiescence, oscillation and depolarization block. In the V - h phase plane, the change of E_{Leak} is reflected in the V -nullcline as moving

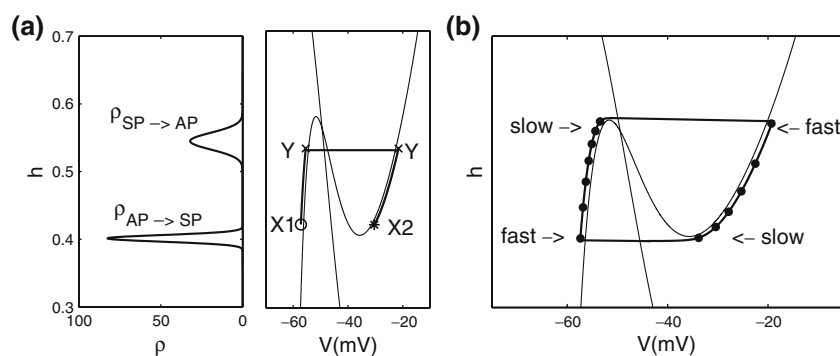


Fig. 5 Transition PDFs and features for estimating temporal correlations. **(a)** Transition PDFs and a trajectory for the SP and following AP in the V - h phase plane. The SP terminates and next AP starts at the same level of h , denoted by Y . Locations of h where the SP starts and next AP ends are denoted by X_1 and X_2 . **(b)** The relative speed of a system in each state. One cycle of the noise-free trajectory

is plotted in the phase plane. Filled circles correspond to equal time intervals (interval in the SP is four times that in the AP) thereby indicating relative speed. When the flow is fast, the dots are widely separated as seen at the beginning of the AP. The system slows down near the termination of each state compared to the initiation because it gets closer to the h or w -nullcline

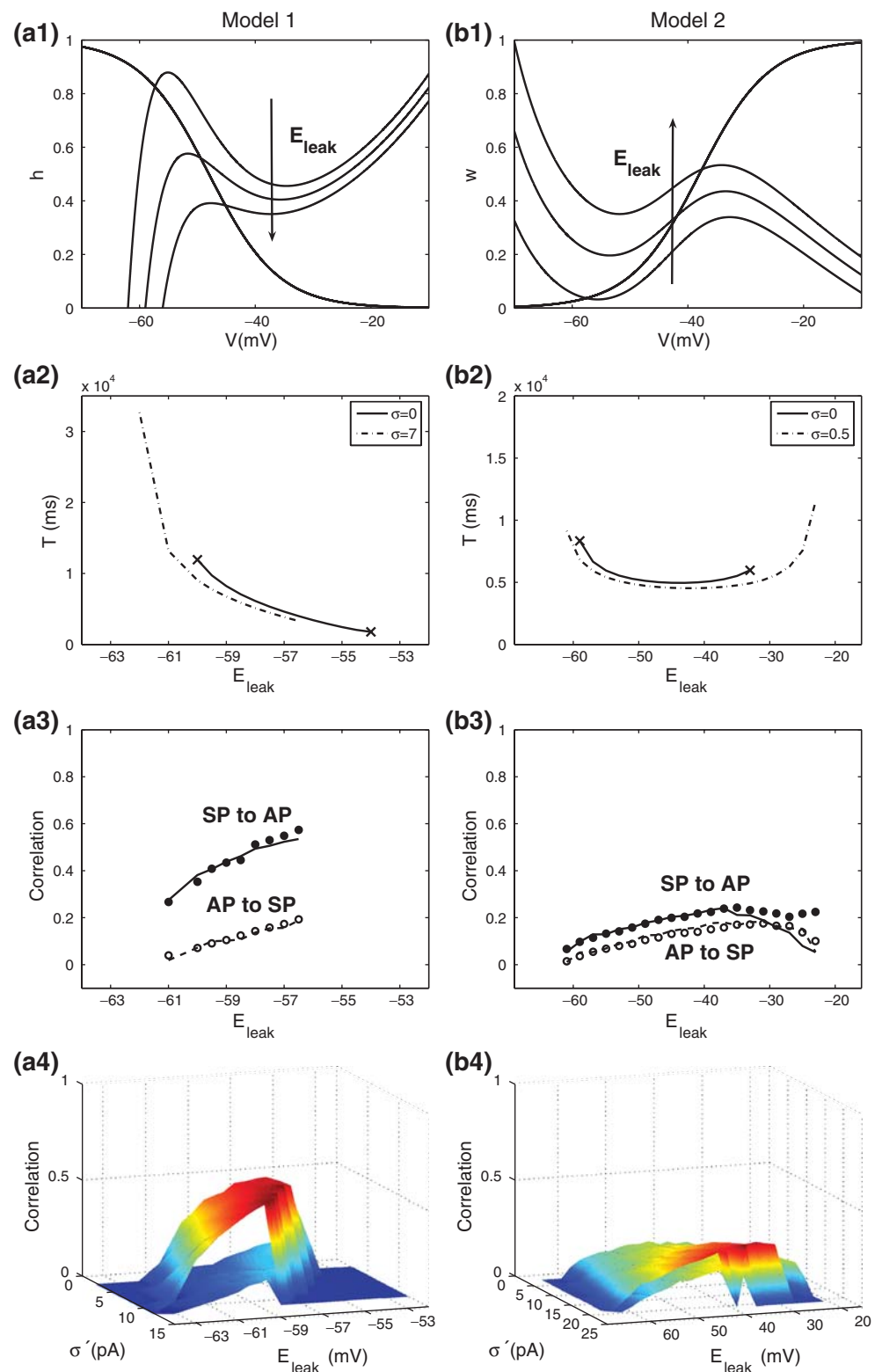
the V -nullcline vertically down in *model 1* and up in *model 2* (Fig. 6(a1 and b1)). With a change of E_{Leak} , the range of the slow variable on the left branch corresponding to the SP varies a lot compared to that of the right branch, the AP in *model 1* (Fig. 6(a1)). On the other hand, the change in the w -ranges for the left and right branches are similar in *model 2* (Fig. 6(b1)). The mean period of oscillation is plotted as E_{Leak} is varied in Fig. 6(a2 and b2). We can see the monotonic dependence of the mean period on E_{Leak} in *model 1*, while the mean period changes non-monotonically as E_{Leak} varies in *model 2*. Noise extends the oscillation regime in both models except high E_{Leak} in *model 1* (Fig. 6(a2)). In Fig. 6(a1), the cubic shape of the V -nullcline becomes flatter with increasing E_{Leak} and two knees of the V -nullcline get closer. So the transition is hard to define under the presence of noise and the oscillation terminates for lower E_{Leak} compared to the deterministic case.

What about the statistical properties in time? Dependence of temporal correlations on parameters, E_{Leak} and σ' is quantified in Fig. 6(a, 3–4) and (b, 3–4). The separation of the two temporal correlations in *model 1* is robust across the different modes of activity and the different noise levels (Fig. 6(a, 3–4)). The temporal correlation between the SP and next AP is larger than that between the AP and next SP even when the system is excitable. In contrast, for *model 2* the temporal correlations are similar over different E_{Leak} and σ' (Fig. 6(b, 3–4)). Note that in both models, the temporal correlations tend to zero as the mean period gets longer. For instance, in the large SP duration case with low E_{Leak} , the system slows down as it approaches the h or w -nullcline and it spends longer time near the knee in the SP. Then the temporal correlation between AP duration and next SP duration is reduced according to Eq. (6f) since the amplification factor of the variability, $\frac{c_{SP}^d}{c_{SP}^s}$ increases. How-

ever, increasing $\frac{c_{SP}^d}{c_{SP}^s}$ in Eq. (6e) doesn't guarantee decrement of the temporal correlation between the SP duration and the following AP-duration. The assumption of constant speed near the transitions breaks down when the speed of motion changes considerably in a small range of h . Consider an extreme case such as an excitable system with very low E_{Leak} in which the system has a fixed point on the left branch of the V -nullcline. The transition from SP to AP occurs at fixed h approximately in a Poisson manner and the variability in the SP duration is totally independent of the jump locations. This is opposite to the underlying idea that jump locations determine the duration of the state. Thus, approximate expressions derived under the assumption of piecewise constant speed in the AP and SP, are good estimates for parameter ranges that are intermediate between the cases of either large AP and large SP durations (Fig. 6(a3 and b3)).

The differences in correlation behaviors between the two models result from the sensitivity of states to current fluctuations. In *model 1*, the V -nullcline's left knee responds more sensitively than the right knee to variation of E_{Leak} (Fig. 6(a1)). This is due to the multiplication of the rapid activation of I_{NaP} which is almost 0 in the SP. To compensate the current change and to overcome multiplication of a number close to 0, the slow variable in the SP needs to change a lot compared to that in the AP. This can be understood as different sensitivity of the states to current change: the SP is more sensitive than the AP in *model 1*. On the other hand, for *model 2* the changes of the left and right branches are similar and thus, both states have similar sensitivities (Fig. 6(b1)). Let's suppose the SP and AP durations are comparable. Then, when the SP is more sensitive to noise than the AP, transitions from the SP are highly random and h values for switching vary widely, while noise has little effect on where the transition from the

Fig. 6 Dependence of dynamics and statistical properties on mean excitability E_{Leak} and the amplitude of noise σ' . *Left: (a, 1–4) model 1. Right: (b, 1–4) model 2.* Panel 1 shows the change of the V -nullcline as E_{Leak} increases and panel 2 plots the mean period of the limit cycle (if it exists) with and without noise. In *model 1*, σ' is set to 7pA (a2) and in *model 2*, it is 10.5pA (b2). In a deterministic system, a limit cycle appears and terminates through a subcritical Hopf bifurcation except that the termination occurs via a supercritical Hopf in *model 2*. The period decreases monotonically in *model 1* as E_{Leak} increases while the dependence of the period on E_{Leak} is non-monotonic in *model 2*. Panels in rows 3 and 4 show the temporal correlations of successive states depending on E_{Leak} and σ' . The solid dots and open circles in a3 and b3 are the temporal correlations from the approximate analysis using Eq. (6e) and (6f). The variances of transition PDFs and amplification factors are obtained from the simulation. The speeds on the slow manifolds at the peak of the transition PDFs are chosen as the constant speeds at the transitions, $c_{\text{SP}}^I, c_{\text{SP}}^T, c_{\text{AP}}^I$ and c_{AP}^T (when the peak is off the knee, then the speed at the knee is used.) Over a broad range of parameter values, temporal correlations are substantially different in *model 1* compared to *model 2*



AP to SP occurs. Therefore, with different sensitivities to noise of the two branches, variances of the transition PDFs and temporal correlations are very different in *model 1* away from the cases of large SP durations like the extreme

case discussed before. If the duration of one state is considerably longer than that of the other state, then the variance of a transition PDF can be reduced even if the state is highly sensitive to noise.

3.5 Relative sensitivities of the AP and SP to different feedback mechanisms

In Section 3.4, we described the dependence of some statistical properties on the underlying mechanisms. Different negative feedback mechanisms provide different patterns of relative sensitivities of the AP and SP to noise and in result, affect the transition PDFs and temporal correlations. Dependence of sensitivity to noise and of the transition PDF's variance on the underlying mechanism can be explained semi-quantitatively. If the noise is small, then the transition happens in a small range of the slow variable. Asymptotically, we can express the variance of the transition PDF as a function of the noise amplitude σ and the differential of the potential depth near the transition as follows (DeVill et al. 2006). Let y denote the level of the slow variable where the transition occurs and y^* is the prediction of y from the theory. Then, in the particular limit, as σ tends to zero and y becomes slower (i.e., the time constant of y , τ_y , increases) such that $\frac{1}{2}\sigma^2 \ln \tau_y$ remains constant, we have:

$$\sqrt{y^2 - \langle y \rangle^2} = \frac{\pi\sigma^2}{2\sqrt{6} \left| \frac{d\Delta U(y^*)}{dy} \right|} \quad (7)$$

Sensitivity to noise is determined by the underlying dynamics, since in the following formula, the variance of the transition PDF is inversely proportional to the differential of potential well along the slow manifold and the potential is the integral of the instantaneous current-voltage relation.

Model 1

$$\Delta U(h) = \int_{V_1(h)}^{V_2(h)} \bar{g}_{NaP} m_{P,\infty}(V) h (V - E_{Na}) + g_{Leak} (V - E_{Leak}) dV \quad (8a)$$

$$\frac{d\Delta U(h)}{dh} = \int_{V_1(h)}^{V_2(h)} \bar{g}_{NaP} m_{P,\infty}(V) (V - E_{Na}) dV \quad (8b)$$

Model 2

$$\Delta U(w) = \int_{V_1(w)}^{V_2(w)} \bar{g}_{NaP} m_{P,\infty}(V) (V - E_{Na}) + \bar{g}_{KS} w (V - E_K) + g_{Leak} (V - E_{Leak}) dV \quad (9a)$$

$$\frac{d\Delta U(w)}{dw} = \int_{V_1(w)}^{V_2(w)} \bar{g}_{KS} (V - E_K) dV \quad (9b)$$

In *model 1*, multiplication of the rapid activating gating variable makes the differential of the potential well to be asymmetric and the SP to be more sensitive to noise than the AP. In *model 2*, the slow activation of I_{KS} is not multiplied by asymmetric values such as a voltage-dependent activating function in *model 1*, and sensitivities of two states are not significantly different.

4 Discussion

We have analyzed noise-induced transitions between the active phase (AP) and silent phase (SP) in neuronal pacemaker models with relaxation dynamics. Dissection into a fast-slow dynamics enables us to describe the dynamics with a slowly changing double-well potential and to characterize noise-induced transitions using Kramers' theory and survival analysis. Our analytic description of the distributions of transition points and durations of AP and SP agrees well with simulated results. We further show that sensitivities to noise in the AP and SP and the relative speed of the slow process determine the temporal correlations of successive states.

We applied our analysis to reduced models of bursting neurons in the preBötC, while considering two different mechanisms for slow negative feedback: either, slow inactivation of an inward current (*model 1*) or slow activation of an outward current (*model 2*) (Butera et al. 1999). Noise affects differently the transition statistics for these two implementations. In *model 2*, the AP and SP are equally sensitive to noise and successive durations show similar, in fact little, temporal correlations. In contrast, *model 1* responds more sensitively to noise in the SP than in the AP and it exhibits significant temporal correlation between SP-duration and next AP-duration but not vice versa, AP to next SP. One may intuitively expect that high sensitivity of the SP state will lead to the stronger temporal correlation, SP to AP. In fact, we showed by assuming constant velocity within a state just this, that the temporal correlations are expressed simply as fractional ratios of the variances of transition PDFs. Although this approximation is useful for qualitative understanding, the expected temporal correlations can be reduced by the non-constant speed of motion. By accounting for the difference in speed of motion at the initiation and termination of the states, we obtained correction factors for fractional ratios and very good quantitative estimates for the temporal correlations. The differences that we find between the temporal correlations for the two feedback mechanisms in the model suggest that one could potentially obtain experimental support for one mechanism over the other (an on-going consideration in this system).

4.1 Noisy oscillator or bistability with noise-induced switching?

As shown in the previous sections, noise may lead to random switching behavior even when the noise-free system is excitable or bistable. Some neuronal models implement noise as the main cause for such oscillatory behavior. For instance, neuronal competition models account for the perceptual switches when ambiguous stimuli are presented. What causes the switches remains an open question. In one framework, slow negative feedback generates periodic alternations in dominance and noise perturbs the regularity, which is called a noisy oscillator model (Laing and Chow 2002; Shpiro et al. 2007). In a different framework such as an attractor network model, slow negative feedback is weak and switches are induced by noise (Moreno-Bote et al. 2007). A recent study showed that statistics of the perceptual switches from the experiments such as the mean period and the coefficient of variation impose constraints on the model's parameters to lie near the boundary defined by the two mechanisms (Shprio et al. 2009).

Insight can be gained into how noise can change the dynamics substantially through its interplay with the slow process by considering the case of relaxation dynamics. The more slowly the dynamics evolves, the more time is allowed for noise to act, and it advances the transition. On the other hand, as noise gets smaller, the premature jump is harder to make and the trajectory becomes similar to the deterministic case. Thus, the noise level and the slow time scale counteract to set the transition location and the mean period of oscillation. If the noise level and slow time scale are balanced appropriately, very regular oscillatory behavior can be induced in a system that, when noise-free, is non-oscillatory (Freidlin 2001; DeVille et al. 2005; Muratov et al. 2005). This phenomenon, called self-induced stochastic resonance (SISR), occurs and was analyzed asymptotically as the noise and the rate of the slow variable tend to zero in a particular relationship. As the noise gets smaller and the time scale gets slower the variability of the transition levels is reduced. In the limit the transitions occur at particular values of the slow variable. When the parameters are away from the limit, the dependence of the statistical quantities can be investigated using analytical expressions for the transition PDFs as we have obtained in this paper. See also the extension for the general case for small noise and slow time constant (DeVile et al. 2006).

The fact that noise may induce (random) oscillations outside the parameter regime for the deterministic system means that noise increases the system's dynamic range. Rhythmicity may occur for a wider range of (steady) stimulus values due to noise. This is well known for spiking neuron models, e.g. (Gutkin and Ermentrout 1998) although less so for slow wave systems.

4.2 Noise-induced bursting—in consideration of spikes during the active phase

Including the spikes We have analyzed noise-induced transitions in reduced versions of bursting models; models in which spike-generating currents were disabled. Suppose we employ a two-variable Hodgkin-Huxley-like mechanism for generating spikes during the AP, as in square-wave bursting (Butera et al. 1999; Rinzel 1987). The time course in Fig. 7(a1) is from *model 1* with the spike-generating mechanism restored. The phase plane projection shows significant variability in the AP→SP transition levels for h , more variability than for the SP→AP transition levels; (Fig. 7(a2)). The temporal correlation from AP to SP is larger than from SP to AP—just opposite to the slow wave case (Fig. 7(a3), compare with Fig. 6(a3)).

What about analysis? In this case our fast subsystem would be two-dimensional and we would retain a single slow variable, h or w . If we describe the escape rate for a fixed slow variable, the transition PDFs can be obtained using survival analysis. However, the fast subsystem is no longer a gradient system; the escape rate cannot be obtained from Kramers' rate. In such a more general system, the escape rate under the presence of noise is described by Wentzell-Freidlin theory (Freidlin and Wentzell 1998). The mean escape rate is exponentially dependent on the noise amplitude and on a quasi-potential that is analogous to the potential depth in Kramers' theory. The quasi-potential can be interpreted as the minimum energy to escape from the attracting domain. But to calculate the quasi-potential for an escaping event from a limit cycle is problematic, since points on a limit cycle are equipotential and the escape path is not unique. Another approach to describe the statistics of escaping events in relaxation dynamics with spikes is to include the effects of noise on the Poincare map of a limit cycle (Hitezenko and Medvedev 2009). We will be exploring these alternatives in future work.

Precluding the spikes—an experimental opportunity How do we apply our slow-wave-based analysis to a burster? What experiments would we propose? One may attempt to convert a bursting cell to a slow-wave pacemaker by pharmacologically blocking some spike-generating currents and thereby hope to explore temporal correlations. The discriminatory value of such an experiment is problematic because blocking agents may not be adequately selective. For example, using tetrodotoxin to block the sodium current for spike-generation will also interfere with the persistent sodium current and thereby the slow negative feedback. Even if the manipulated neuron shows pacemaking slow waves the underlying mechanism would be compromised if *model 1* applied to the bursting case.

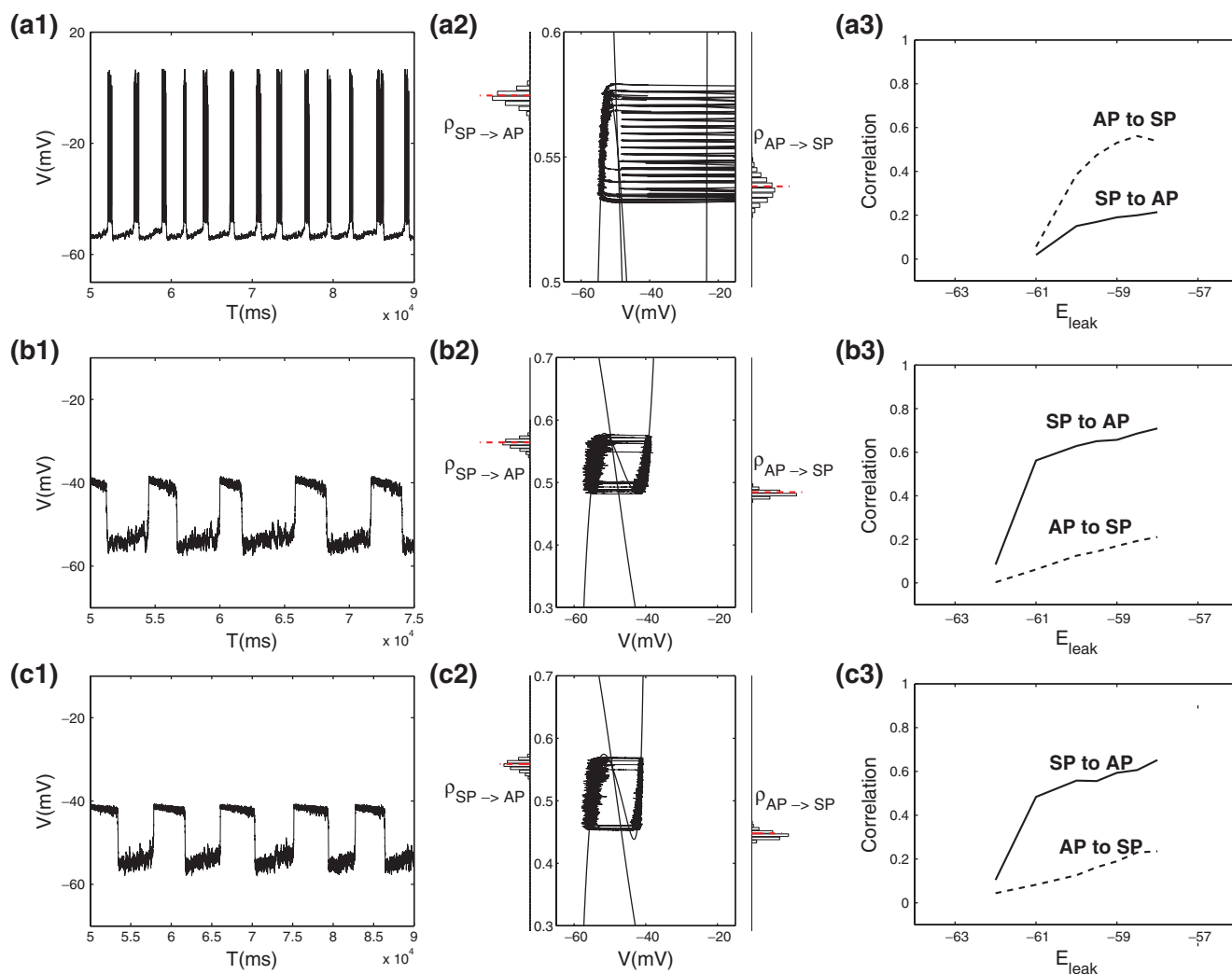


Fig. 7 Time course and statistical properties for model 1 including spikes (row *a*) and shunting spikes via dynamic clamping (rows *b* and *c*). The noise level σ' is set to 2 pA (*a*) and 7 pA (*b* and *c*). In *b* and *c*, the spikes are precluded by including an instantaneously- V -gated current that activates in the voltage range of the *AP*. This current can be either an inward current, $I_{\text{inward}}(V) = 10(V - E_{\text{Na}})/[1 + \exp(-0.5V - 17)]$, that is subtracted (*b*,1–3) or an outward current, $I_{\text{outward}}(V) = 20(V - E_K)/[1 + \exp(-V - 38)]$, that is added (*c*,1–3). Column 1 shows the time course of the voltage. For the control case (*a*1), I_{shunt} is absent; the *AP* shows normal spiking. In *b*1 and *c*1, the spike-generating currents are still present but the system generates the slow wave

oscillation because I_{shunt} overwhelms or shunts the spike generating currents. Column 2 shows the trajectory in the projected V - h phase plane with the transition *PDFs* (the spikes are clipped at -10 mV in *a*1) and column 3 shows the temporal correlations for different E_{Leak} . In contrast to the slow wave oscillation, the levels of h for the bursting pattern (*a*2) are more variable in the *AP* than in the *SP* and the temporal correlation from *AP* to *SP* is larger than from *SP* to *AP* (*a*3). By precluding spikes with an instantaneously- V -gated current, the model produces a similar pattern of temporal correlations that resemble those of model 1 as a slow wave system (*b*,2–3 and *c*,2–3)

We propose an alternative approach which does not depend on pharmacology. Suppose we inject into the neuron, with the dynamic-clamp method (Prinz et al. 2004; Sharp et al. 1993) an instantaneously- V -gated current that activates in the voltage range of the *AP* and which overwhelms the spike-generating mechanism. This current (call it $I_{\text{shunt}}(V)$) would not interfere with the *SP* dynamics. The goal is to replace the original *AP* branch that includes spiking by an *AP* branch that is stable, on the fast time scale. Note, we are not proposing to introduce a subtractive virtual current that counteracts spikes on an individual basis; rather we are introducing more of a shunting

effect. $I_{\text{shunt}}(V)$ could be “designed” as an inward current that is subtracted (Fig. 7(b,1–3)) or as an outward current that is added (Fig. 7(c,1–3)). If the parameters for either choice of I_{shunt} are chosen in an appropriate range the neuron would show asymmetries in the *PDFs* for switching values and in temporal correlations as in our slow-wave pacemaker model 1. Correspondingly, when we apply such a recipe to a burster that is well-described by model 2 the *PDFs* and temporal correlations are more similar (small) rather than different or asymmetrical (not shown). Note that the mean *SP* duration is similar to the mean *AP* duration in model 1

(Fig. 7(b1 and c1)). We confirm again that the temporal correlation reflects the underlying slow process and cannot be inferred only from the distributions of the slow variable at transitions between states. We will further elaborate on this procedure in future work.

4.3 Synaptic conductance noise

In the previous sections, noise was modeled as additive white noise current injection. Here, we discuss some implications if noise is due to random synaptic conductance transients, as originating from the independent activity of other neurons. The postsynaptic conductance can be modeled with an exponential decay from a step onset. The time scales of decay can range from a few to tens of msec for excitation and inhibition and the net synaptic conductance is auto-correlated with similar temporal orders. However, since auto-correlation decays much faster than the time scale of the slow variable, we may expect that the auto-correlation of the noise has only modest effect on the noise-induced transitions.

Compared to the additive current injection noise, the synaptic current from conductance noise is activity-dependent. Since the conductance is multiplied by the driving force which is the difference of the membrane potential and the synaptic reversal potential, the synaptic current will differ depending on the activity state. For a given relative weighting between the excitatory and inhibitory conductance inputs the amplitudes of total synaptic current in the *SP* and *AP* will differ. For instance, the inhibitory synaptic current has larger amplitude during the *AP* compared to the *SP*. As we increase the portion of the excitatory conductance input, the size of the synaptic current in the *SP* increases more than that in the *AP*. Also, the relative contribution of synaptic conductance to total membrane conductance will differ between *SP* and *AP*, since the intrinsic conductances are strongly voltage-dependent.

In Eq. (7), the amplitude of noise determines the variances of transition *PDFs* as well as the underlying mechanisms. As discussed in Sections 3.3 and 3.4, the variances of transition *PDFs* are crucial for temporal correlations. Thus, the synaptic conductance noise with different balance between the excitatory and inhibitory inputs may allow various patterns of temporal correlations. However, the mean synaptic current noise is not balanced (at zero) as time evolves. The mean will alternate slowly and can change the dynamics during the *SP* and *AP* differently. So the effect of the mean synaptic input should be factored in when we consider the switching statistics.

4.4 Application to a neuronal network model for slow episodic behavior

We have considered statistical properties of a cellular pacemaker neuron with noise. Our framework can be

extended to analyze statistical properties of network activity having similar relaxation feature. Spontaneous episodic activity occurs in developing neuronal networks (Grzywacz and Sernagor 2000; Tabak et al. 2000). Mean field-like, firing rate, network models for a single recurrent excitatory population show similar slow wave episodic rhythms with sharp transitions between an *AP* and *SP* (Tabak et al. 2000). For these models, as for the cellular pacemaker case, alternative mechanisms for the slow negative feedback have been introduced: synaptic depression (*s*-model) and cellular or spike-frequency adaptation (*θ*-model).

Synaptic depression slowly modulates the regenerative excitatory feedback, in analogy to inactivation of an inward current for the cellular model. It is referred to as divisive negative feedback. On the other hand, cellular adaptation is formulated as a slowly growing negative term in the net input, analogous to activation of outward current; it is called subtractive negative feedback. Thus, the *s*-model and *θ*-model respond to change of an input control parameter in a similar way as *model 1* and *model 2* for, say, changing E_{Leak} . Similarity of the developing network models and cellular pacemaker models with different feedback mechanisms has been discussed in terms of the control of the *SP* and *AP* durations by the parameters (Tabak et al. 2006).

As we showed for the cellular model, additive noise in the *s*-model for the divisive negative feedback leads to asymmetrical sensitivities for transitions, *AP* to *SP* and *SP* to *AP*, in contrast to more symmetrical effects for the *θ*-model for the subtractive feedback (Tabak et al. SFN poster 2007). We have confirmed that the temporal correlation from the *SP* to *AP* is substantially larger than that from the *AP* to *SP* over a broad range of input in the *s*-model, while they are similar in the *θ*-model (not shown here). For parameter regimes where the speed of slow motion doesn't change much in each state, our approximate analysis of temporal correlations in Eq. (6d) and (6e) can be applied. As true for the cellular pacemaker neuron model, the effect of the slow negative feedback, in the divisive mechanism, is strongly dependent on the population activity. This induces the asymmetry in sensitivities for transitions and temporal correlations between active and silent states. We conclude that the differences in statistical properties allow us to distinguish among biophysical mechanisms: substantially different temporal correlations (larger for *SP* to *AP* than for *AP* to *SP*) are expected with the divisive feedback while similar temporal correlations are expected with the subtractive feedback.

5 Conclusions

Both noise and slow negative feedback contribute to the random switching between activity states in many neuronal

slow wave systems. One expects that the slow feedback leads to correlations between durations of successive states. For example, incomplete recovery because of premature termination of a silent phase will lead a shorter next active phase. The sensitivity to noise and perturbations for causing a switch from a state affects the probability for terminating a switch. Our results here, from fast–slow analysis in the case of stochastic relaxation dynamics, emphasize that the distribution of termination points (recovery levels) depends also on the speed of the slow feedback process. We derived these probability density functions based on Kramers rate and survival analysis. However, just knowing these distributions is not enough to predict the temporal correlations—the variation in speed along the slowly varying state also enters. We provide formulae for the temporal correlations that include both effects: the variability in recovery rates and relative speeds within a state. These formulae are fractional speed-weighted variance ratios. We have emphasized by way of example, using a cellular pacemaker model (Butera et al. 1999), that asymmetries in temporal correlations, *SP* to *AP* differing substantially from *AP* to *SP*, may provide a criterion by which to distinguish qualitatively different feedback mechanisms—either subtractive/additive or divisive/multiplicative. In applications where the mechanistic basis is unsettled these second-order statistics may provide evidence in favor of one mechanism over the other. We expect that our approach is extendable beyond the single-fast and single-slow variable framework that we considered here to, say, multi-population models for alternations in neuronal competition formulations.

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Appendix

Modification of the duration

In Section 3.2, the durations of states are computed under the assumption that the system moves along the V -nullcline. However, as seen in Fig. 3(b), the trajectory during each state is widely spread along the V -nullcline due to noise. The deviation of the trajectory from the V -nullcline affects the speed of the slow variable in Eq. (3c) and thus, may lead to the discrepancy between the analysis and the simulation (Fig. 3(d)).

The statistics of this deviation can be obtained analytically by the stationary distribution of V for each h . The distribution of V at each h is the stationary solution for the Fokker-Planck equation (Gardiner 1985). When the system has a potential function, $U(V, h)$, then the stationary solution is given as follows:

$$p_s(V, h) = C \exp(-2U(V, h)/\sigma^2) \quad (10a)$$

In a double-well potential, we can calculate the conditional probability and the mean given that the system stays in the left well by normalizing Eq. (10a). Here, $V_{sep}(h)$ is the separatrix between the left and right wells for given h .

$$p_{cond}(V, h) = \frac{p_s(V, h)}{\int_{-\infty}^{V_{sep}(h)} p_s(V', h) dV'} \quad (10b)$$

$$\mu(h) = \int_{-\infty}^{V_{sep}(h)} p_{cond}(V', h) V' dV' \quad (10c)$$

With the modified V obtained from Eq. (10c), we can compute the duration. Figure 8 shows the mean deviation

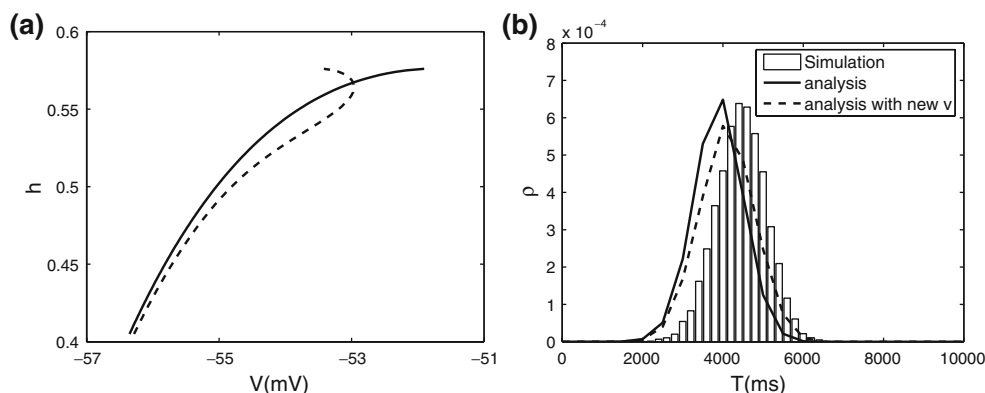


Fig. 8 Deviation of a trajectory from the slow manifold and its effect on the duration of the state. (a) Mean V of the noisy trajectory (dotted line) and the V -nullcline in the SP (solid line) are plotted in the V - h phase plane. Due to noise, the mean of the trajectory deviates from the V -nullcline. (b) Distribution of SP durations obtained from a

simulation and an analysis. In an analysis, the distributions of starting and leaving locations in the SP are given as the transition PDFs in h obtained from the simulation. The solid line is computed with the V values on the V -nullcline and the dotted line is computed using the mean of the deviated V

of the trajectory from the V -nullcline and comparison of the durations computed using the V -values on the V -nullcline and the modified V . The distributions of h at the entry and leaving points are obtained from the simulations so that the discrepancy between the simulation and analysis does not originate from the transition $PDFs$. With the modified V , there is a modest change in the distribution of durations. It agrees better with the simulation even though the distribution with the modified V still doesn't agree with the simulation perfectly. The mean of the deviation is not enough to correct the distribution. However, we can see the influence of the deviation of V on the duration in Fig. 8(b).

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